

Question number	Answer	Mark
6(a)	An answer that includes the following point: B 1 is the correct answer • A, 0 is not the correct answer as the third statement only is correct • C, 2 is not the correct answer as the third statement only is correct • D, 3 is not the correct answer as the third statement only is correct	

Question number	Answer	Additional guidance	Mark
6(b)(i)	A calculation showing the following steps: • calculation of cardiac output / min (1) • conversion to dm^3 and output per hour and given to two significant figures (1)	$(77 \times 70) = 5390$ $((5390 \times 60) / 1000) = 320$ Correct answer gains 2 marks x60 first and do the x70 afterwards. e.g. 1 mark for 4620 if they don't get the correct answer	(2)

Question number	Answer	Additional guidance	Mark
6(b)(ii)	A calculation showing the following steps: • calculation of cardiac output (1) • correct unit stated (1) • calculation of cardiac index (1)	mp1 for $(1.49 \times 2.8) = 4.2$ /4.17 / 4.172 mp2 for $\text{dm}^3\text{min}^{-1}\text{m}^{-2}$ ACCEPT $\text{dm}^3 \text{ m}^{-2} \text{ min}^{-1}$ ACCEPT dm^3 per min per m^2 Mp3 for $(6.9/1.92) = 3.5$ / 3.49 / 3.489 ACCEPT values / units in working if table is empty (or by formula)	(3)

Question number	Answer Additional guidance	Mark
6(b)(iii)	An answer that includes the following points: Indicative content - description from graphs and experimental data Heart rate: • as the concentration of adrenaline increases, the increase in heart rate increases in both groups • this increase is (slightly) {greater in the older group of people / lower in younger people} • at all concentrations of adrenaline the older group have greater {change/heart rate} / at all concentrations of adrenaline the younger group have smaller {change/heart rate} • but there are no error bars to judge whether or not this difference is significant • due to stated effect of aging on heart – weaker muscle/effect on SAN/lower stroke volume Cardiac index: • as the concentration of adrenaline increases, the increase in cardiac index increases in both groups • the increase is the same at {20 / 40} ng kg⁻¹ min⁻¹ • at concentrations above 40 ng kg⁻¹ min⁻¹ the increase is {lower in older people / greater in younger people} • overall the increase cardiac index is {lower for the older people / greater for younger people} • due to lower cardiac output / increased body surface area/lower stroke volume / (accept converse for younger people) • but there are no error bars to judge whether or not this difference is significant	(6)

Change in blood pressure

- blood pressures decrease and then increase in both groups (as adrenaline concentration increases)
- a concentration of $20 \text{ ng kg}^{-1} \text{ min}^{-1}$ decreases the blood pressure in both groups
- a concentration of **{84-120} $\text{ng kg}^{-1} \text{ min}^{-1}$** increases the blood pressure in older people
- in older people lower change in blood pressure (in response to higher concentrations of adrenaline) / converse for younger people

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- due to stated changes of aging on heart muscle/ blood vessels/baroreceptors
 - no overlap of error bars between older and younger people indicate that these differences are significant (at all concentrations of adrenaline) / the error bars overlap for some concentrations for older people meaning that these differences are not significant / the error bars do not overlap for some concentrations for younger people meaning that these differences are significant

Level 1:

1 or 2 graphs commented on changes in heart rate/cardiac index/blood pressure as concentration of adrenalin increases

Level 2:

2 or 3 graphs commented on, including details of the effect of adrenaline on older people

Level 3:

All 3 graphs commented on, including details of the effect of adrenaline on older people and discussion of the significance of the data.